

Reinforcement Learning Benchmarking For ViZDOOM: Target Acquisition & Movement Prediction

A Model Comparison

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Abstract— This paper presents a reinforcement learning (RL) experimentation suite designed for the ViZDoom Defend-the-Line scenario, enabling reproducible, and scalable benchmarking of RL agents in the task of defending a position from multiple assailants. The system integrates value-based (DQN), exploration-enhanced (NoisyNet-DQN), and policy-gradient (Advantage Actor-Critic, A2C) architectures into a shared experimental pipeline with standardized preprocessing, reward shaping, model persistence, metric logging, and video-based behavioral inspection. The framework supports systematic comparisons of learning stability, exploration efficiency, and tactical decision-making in visually rich first-person environments. Results highlight distinct performance profiles across agents and demonstrate the importance of structured exploration, n-step bootstrapping, and shaping-informed reward engineering. The discussed experiment and framework are positioned as a foundation for future studies by integrating more advanced architectural extensions to assess capacity for target acquisition and target tracking.

Keywords— Value-based learning, Policy-gradient learning, Machine vision, Target acquisition, Strategic decision-making

The development of deep reinforcement learning has progressed through several transformative milestones, each expanding the scope of RL from controlled grid environments to high-dimensional, vision-based domains such as ViZDoom. Early RL agents relied on tabular methods and handcrafted features, limiting scalability to small state spaces. A major leap occurred with the introduction of the Deep Q-Network (DQN) by Mnih et al. [1], which demonstrated that convolutional neural networks could learn effective policies directly from raw pixel inputs in the Atari 2600 benchmark suite. This work established deep RL as a powerful paradigm for visual decision-making and sparked widespread interest in combining deep learning with temporal-difference methods for complex tasks.

Subsequent architectures addressed the limitations of early DQN agents. Techniques such as Double DQN, Dueling Networks, and Prioritized Experience Replay[2-4] were introduced to mitigate overestimation bias, improve state-action discrimination, and increase sample efficiency. Parallel

research explored stochastic exploration mechanisms, leading to Noisy Networks [5], which inject parametric noise into network weights to produce structured exploration behavior superior to ϵ -greedy randomness.

In parallel, policy-gradient and actor-critic methods advanced significantly. Algorithms such as A3C, A2C, PPO, and SAC [6-7] demonstrated improved stability, reduced gradient variance, and better adaptability to continuous control. These developments expanded the applicability of RL beyond Atari-like tasks into more dynamic, physics-rich environments such as MuJoCo, Unity ML-Agents, and real-time strategy simulators. More recent trends include the integration of transformers, world models, and model-based planning, enabling long-horizon reasoning and improved sample efficiency through predictive representations.

Within this historical context, ViZDoom occupies an important niche as a visually rich, partially observable, adversarial environment that stresses real-time reaction and target prioritization. By situating DQN, NoisyNet-DQN, and A2C within the evolution of RL research, the present benchmarking suite provides a controlled experimental setting for examining how classical and modern RL innovations perform under FPS-style constraints.

I. RELATED WORK

Deep Q-Networks (DQN) demonstrated the first major success of value-based deep RL in Atari-like environments, motivating subsequent FPS applications. Extensions such as Noisy Networks introduced parametric noise to promote structured exploration, addressing the shortcomings of ϵ -greedy policies. Actor-critic methods, including A2C and its asynchronous variants, provide smoother gradient updates by leveraging advantage estimation and parallel rollouts.

Reward shaping is frequently explored in sparse-reward environments to accelerate convergence, but requires careful calibration to avoid distorting optimal policies. This work integrates configurable shaping policies to enable fine-grained ablation studies that examine their effects on early- and late-stage policy development

TABLE 1: Miniature Literature Matrix

Citation (short)	Agent / Method	Key Contribution / Focus
Mnih et al., 2015, <i>Nature</i> (Nature)[1]	Deep Q-Network (DQN)	First major deep RL agent to achieve human-level control from pixels using CNNs + replay + target network. Establishes Atari as a standard video-game benchmark.
Lample & Chaplot, 2017, AAAI (arXiv)[8]	DRQN + auxiliary tasks (e.g., object detection)	Demonstrates deep RL agents playing first-person shooters , using recurrent models and auxiliary tasks to handle partial observability and memory.
Fortunato et al., 2018, ICLR (NoisyNet) (arXiv)[5]	DQN variants with Noisy Networks	Introduces NoisyNet exploration by adding parametric noise to weights; shows improved performance across 57 Atari games vs. ϵ -greedy exploration.
Badia et al., 2020, <i>Agent57</i> (arXiv)[9]	Agent57 (meta-controller + exploration)	First agent to outperform humans on all 57 Atari games , combining exploration strategies and a meta-controller to manage timescales and curiosity.
Berner et al., 2019, OpenAI Five (arXiv)[10]	Policy-gradient with large-scale self-play (OpenAI Five)	Shows that large-scale self-play RL can reach superhuman performance in Dota 2, using huge compute and continual training.

II. PROBLEM

Training autonomous agents to operate effectively in visually rich first-person environments poses a fundamental challenge in reinforcement learning due to partial observability, stochastic enemy behavior, and the need to balance precision with rapid decision-making. In the ViZDoom Defend-the-Line scenario, an agent must identify and eliminate approaching enemies while minimizing unnecessary actions and avoiding damage—tasks that require accurate perception, efficient exploration, and stable policy learning from raw pixel observations.

Existing RL methods vary significantly in how they represent value, assign credit, and explore high-dimensional action spaces, leading to divergent behavioral tendencies such as over-aggressive firing, poor temporal credit assignment, or overly conservative policies. The central problem addressed in this work is therefore to systematically evaluate how different reinforcement learning architectures—specifically a Baseline DQN, a NoisyNet-DQN, and an A2C agent—learn tactical behavior in a constrained FPS environment, and to determine how architectural choices, exploration mechanisms, and reward-shaping strategies influence accuracy, stability, and overall task performance.

By establishing a consistent experimental framework and shared evaluation protocol, this study aims to clarify which algorithmic properties are most conducive to learning effective combat behavior in complex visual domains.

III. ENVIRONMENT

The experiments are conducted in the ViZDoom Defend-the-Line scenario, a constrained first-person reinforcement learning environment defined by a partially observable Markov decision process (POMDP). At each timestep t , the agent observes a raw pixel frame $s_t \in \mathbb{R}^{1 \times 30 \times 45}$, representing a downsampled grayscale image of the forward-facing camera. The agent does not possess explicit information about enemy positions, velocities, or distances; instead, it must infer these properties solely through visual perception.

The action space is discrete and consists of three primitive actions: $\{\text{TURN_LEFT}, \text{TURN_RIGHT}, \text{ATTACK}\}$, optionally combined into composite commands by the environment’s action builder. The scenario restricts locomotion, fixing the agent’s spatial position while allowing only rotational adjustments and firing. Enemies spawn at varying angles and advance toward the agent; failure to eliminate them before they reach melee or firing distance results in health loss and may terminate the episode prematurely.

The reward structure is designed to capture the competing objectives of effective threat neutralization and firing efficiency. Let $k_t \in \{0,1\}$ indicate whether an enemy is eliminated at timestep t , and let $s_t^{\text{shot}} \in \{0,1\}$ denote whether the agent fires a shot. Additionally, let

$$\alpha = 0.2, \beta = 0.01$$

represent the accuracy bonus scale and shot penalty scale, respectively. Define cumulative kills and shots up to timestep t as

$$K_t = \sum_{\tau=1}^t k_{\tau}, S_t = \sum_{\tau=1}^t s_{\tau}. \quad (1)$$

The instantaneous reward is then given by

$$r_t = k_t + \alpha \cdot \frac{K_t}{S_t + \varepsilon} - \beta \cdot s_t, \quad (2)$$

where $\varepsilon > 0$ is a small constant to avoid division by zero.

The first term encourages successful eliminations, the second term rewards sustained firing accuracy, and the third penalizes excessive or indiscriminate firing. For value-based agents, multi-step returns are computed using an n -step horizon of $n=3$, producing shaped returns

$$R_t^{(n)} = \sum_{i=0}^{n-1} \gamma^i r_{t+i} + \gamma^n V(s_{t+n}), \quad (3)$$

with discount factor $\gamma=0.99$. Actor-critic methods instead use these rewards within generalized advantage estimation.

An episode terminates when the agent’s health is depleted or the scenario time limit is reached. The combination of sparse lethal threats, continuous visual noise, and firing efficiency constraints yields a challenging decision-making environment that stresses perceptual acuity, temporal credit assignment, and the trade-off between aggression and precision—qualities that differ substantially across the DQN, NoisyNet-DQN, and A2C learning paradigms.

IV. SYSTEM ARCHITECTURE

The RL suite is implemented in a modular structure consisting of environment interfaces, agent constructors, training supervisors, evaluation utilities, and metric logging modules. Each agent interacts with a consistent preprocessing function, a fixed action space, and a standardized ViZDoom scenario, ensuring fairness across comparisons.

A. Overview

All agent evaluations in the framework rely on a system that ensure methodological consistency and reproducibility across experiments. A standardized environment interface initializes the ViZDoom scenario, constructs the discrete action space, and applies a common preprocessing function to transform raw screen buffers into compact tensor inputs. Core hyperparameters—including learning rate, discount factor, steps per epoch, frame repeat, and batch size—are centrally defined to ensure parity across models, while a configurable reward-shaping module introduces optional accuracy bonuses, shot penalties, and n-step return settings. The evaluation pipeline incorporates a shared test_agent procedure that executes multiple episodes per agent, applies identical action-selection logic, and computes summary statistics such as mean reward, variance, and episode-level performance. Additionally, all agents use a uniform logging infrastructure: an EpochLogger records per-epoch metrics to CSV, NumPy arrays store reward and kill trajectories, and model parameters are checkpointed using PyTorch’s serialization utilities. Visualization and comparison modules further standardize post-training analysis by producing learning curves, accuracy plots, and aggregate performance tables. Together, these shared components form a cohesive evaluation ecosystem that isolates architectural differences between DQN, NoisyNet-DQN, and A2C while maintaining rigorous experimental control.

B. Preprocessing

The preprocessing module transforms raw ViZDoom frames into a compact, learning-efficient representation suitable for convolutional neural networks. Each incoming image is spatially downsampled to a 30×45 resolution using bilinear interpolation to reduce computational overhead while preserving essential spatial structure for tactical decision-making. The resized frame is then cast to 32-bit floating point to ensure numerical compatibility with the learning pipeline, and a channel dimension is added to produce a standardized 1×H×W tensor consistent with PyTorch’s channel-first convention. This lightweight preprocessing step significantly reduces memory consumption and accelerates training iterations while maintaining sufficient visual information for effective policy learning within the ViZDoom environment.

C. Interaction with Action Space

Agents in the framework interact with the ViZDoom environment through an identical, centrally defined discrete action space constructed using the build_actions() utility. This function enumerates all admissible combinations of movement and attack commands available in the scenario and maps them to a compact integer-indexed action set. During training and evaluation, each agent receives a preprocessed visual observation and outputs a single discrete action index, which is uniformly translated to the corresponding ViZDoom action vector before being executed via make_action() with a consistent frame-repeat parameter. By abstracting away environment-specific control signals and enforcing a shared

action representation, the system ensures that all models operate under identical behavioral constraints and are evaluated on equivalent tactical decision-making capabilities. This design isolates algorithmic differences—such as exploration mechanisms or value estimation strategies—while preventing confounds introduced by heterogeneous action implementations, thereby enabling fair and reproducible cross-agent performance comparisons.

D. Summary

Key architectural components include:

- Training loop with epoch-based metric logging.
- Support for n-step returns, GAE, and entropy regularization.
- Automated saving of learned weights for reproducibility.
- Visualization tools for reward curves, kill trajectories, and accuracy trends.
- Optional video recording of top-performing episodes.

This modularity enables seamless integration of new algorithms and facilitates research workflows such as ablation studies and architecture benchmarking.

V. METHODS

Three RL agents were implemented for comparative analysis:

A. Baseline Dueling DQN

The Baseline DQN agent implements a Deep Q-Network formulation, in which a convolutional neural network approximates the action–value function $Q(s,a)$ directly from pixel observations. For each incoming frame, the agent produces a discrete action index that corresponds to a ViZDoom action vector, ensuring deterministic, state-conditioned behavior. Learning follows the standard temporal-difference (TD) paradigm: transitions are stored in an experience replay buffer, and mini-batches are repeatedly sampled to compute TD errors and update the network weights through stochastic gradient descent. A periodically updated target network stabilizes the bootstrapping process by providing fixed Q-value estimates for the duration of each update interval. This reduces oscillatory behavior and divergence tendencies commonly observed in pixel-based RL tasks.

The agent is built on a lightweight dueling convolutional network, designed specifically for low-resolution input from the ViZDoom environment. The architecture begins with two convolutional blocks: the first uses a 3×3 kernel, stride 2, and 8 output channels, followed by BatchNorm2d(8) and ReLU activation; the second applies another 3×3 convolution with 16 channels and stride 2, again followed by batch normalization and ReLU. These layers reduce the preprocessed 1×30×45 frame into a compact spatial embedding of size 16×6×10, which is flattened into a 960-dimensional vector (denoted as FLAT_SIZE). The flattened representation is processed by a fully connected layer with 128 hidden units (HIDDEN_SIZE) and a ReLU activation, forming a compact latent feature vector.

The dueling architecture then splits this latent embedding into two output streams: a value stream producing a single

scalar $V(s)$ and an advantage stream producing $A(s,a)$ for each action. The final Q-values are computed by combining these outputs according to

$$Q(s, a) = V(s) + A(s, a) - \frac{1}{|a|} \sum_{a'} A(s, a'). \quad (1)$$

This reduces redundancy in learning action-specific differences when many states share similar value structure—an important property in first-person shooter tasks. Together, the two convolutional layers, the fully connected layer, and the value/advantage heads form a compact yet expressive architecture well-suited for pixel-based Q-learning in ViZDoom.

Choosing Baseline DQN and as primary value-based agent provides a strong methodological foundation for studying reinforcement learning behavior in visually rich first-person environments such as ViZDoom. The classical Deep Q-Network remains one of the most influential architectures in deep reinforcement learning, establishing the first successful end-to-end pipeline for learning control policies from raw pixels. As such, DQN represents a conceptual baseline that is both computationally feasible and empirically well-understood, making it an ideal reference point for evaluating architectural improvements or environment-specific modifications. Its established training dynamics, replay-based learning, and 1-step temporal-difference updates offer a stable framework within which to analyze reward shaping, action selection behavior, and sample efficiency in an FPS environment.

B. NoisyNet Dueling DQN

The The NoisyNet-DQN model builds upon the Baseline DQN architecture by integrating parameter-space noise into its convolutional and fully connected layers. Instead of relying on an external ϵ -greedy policy to induce stochasticity, the network weights themselves contain adaptive noise variables that are sampled at each forward pass. This produces exploration that is both state-dependent and structured: different states may lead to different noise-induced action selections, and repeated visits to the same state may lead to more consistent exploratory behavior compared to random action selection. Because the noise parameters are learned jointly with the primary network weights, the agent can tune the magnitude and direction of its exploratory behavior based on training progress.

This agent shares the same convolutional and dueling stream structure as the baseline DQN but replaces its fully connected advantage and value layers with NoisyLinear layers, implemented in NoisyDuelQNet. These layers augment each weight parameter w_{ij} with learnable noise variables using the parameterization:

$$w_{ij} = \mu_{ij} + \sigma_{ij} \cdot \epsilon_{ij}, \quad (2)$$

where ϵ_{ij} is zero-mean noise sampled at each forward pass. This creates controlled stochasticity in the Q-value predictions, enabling state-dependent exploration without reliance on ϵ -greedy randomness.

The NoisyNet variant replaces the standard value and advantage output layers with NoisyLinear (HIDDEN_SIZE $\rightarrow$ 1) and NoisyLinear (HIDDEN_SIZE $\rightarrow$ n_actions), respectively. Each NoisyLinear layer maintains learned parameters for both the mean μ and standard deviation σ of its weights and biases, injecting structured noise using the factorized Gaussian mechanism proposed by Fortunato et al.

[5] This noise is resampled on every forward pass during training, enabling state-dependent exploration without ϵ -greedy sampling. Because the convolutional backbone is identical to the baseline model, any improvement observed during experimentation stems exclusively from the noisy exploration mechanism rather than increased representational capacity. This provides a clean, controlled comparison between classical Q-learning and parameter-space exploration.

Noisy DQN complements the baseline by addressing one of the most persistent challenges in deep RL: effective exploration. Many FPS environments—including ViZDoom—exhibit deceptive or sparse reward signals, where naïve ϵ -greedy exploration frequently leads to stagnation or repeated suboptimal behavior. Noisy Networks introduce stochasticity directly into the network parameters, providing structured, state-dependent exploration that evolves throughout training and avoids the pitfalls of manually tuned exploration schedules.

C. A2C

For The A2C agent implements a synchronous Advantage Actor - Critic formulation, in which a convolutional neural network jointly parameterizes both a stochastic policy $\pi(a | s)$ and a state-value function $V(s)$ directly from pixel observations. For each incoming frame, the actor head produces a probability distribution over the discrete ViZDoom action set, from which the agent samples an action, enabling inherently stochastic, exploration-aware behavior. Unlike replay-based value methods, A2C learns from on-policy trajectories collected over short rollouts; each transition contributes to the computation of temporal advantages using generalized advantage estimation (GAE), which stabilizes training by reducing variance in the policy gradient. The critic head is simultaneously trained to regress toward bootstrap-enhanced return estimates, providing a learned baseline that improves gradient efficiency. By updating both policy and value parameters at each rollout and applying entropy regularization to prevent premature policy collapse, the A2C agent achieves smoother learning dynamics and greater stability in visually complex, partially observable environments such as ViZDoom.

The agent employs a shared-backbone actor - critic architecture, structurally similar to the convolutional encoder used in the DQN variants. The network begins with two convolutional blocks identical in configuration: a $1 \rightarrow 8$ conv layer (3×3 , stride 2) with batch normalization and ReLU, followed by a $8 \rightarrow 16$ conv layer (3×3 , stride 2) with batch normalization and ReLU. These layers reduce the visual input to a $16 \times 6 \times 10$ tensor, which is flattened and passed through a Linear(960 $\rightarrow$ 128) fully connected layer with a ReLU activation.

From this shared 128-dimensional feature vector, the architecture branches into two distinct output heads. The policy head applies a Linear(128 $\rightarrow$ n_actions) projection to produce logits for a categorical action distribution, enabling stochastic sampling for exploration. The value head simultaneously computes a scalar state value $V(s)$ using a Linear(128 $\rightarrow$ 1) layer. This dual-output structure allows the network to learn both action probabilities and baseline estimates jointly, supporting advantage computation during the rollout update.

1. Policy head (actor): produces a vector of logits over the discrete action space.
2. Value head (critic): outputs a scalar baseline estimate $V(s)$.

A2C serves as an appropriate comparative candidate to the Baseline and NoisyNet-DQN agents because it represents a fundamentally different learning paradigm—policy-gradient and actor-critic methods—while retaining a network architecture and input modality similar enough to allow fair comparison. Whereas DQN variants rely on off-policy temporal-difference updates, experience replay, and bootstrapped Q-value estimation, A2C optimizes a stochastic policy directly through gradient ascent and learns a value baseline to reduce variance. This distinction allows the study to isolate the effects of deterministic versus stochastic action selection, replay-based versus rollout-based learning, and value-only versus actor-critic optimization. Moreover, A2C’s on-policy updates and entropy-regularized exploration provide a principled alternative to the ϵ -greedy and NoisyNet exploration strategies used by the DQN agents. Evaluating these models side-by-side in the same ViZDoom environment offers insight into how different approaches to exploration, temporal credit assignment, and value estimation shape learning behavior in visually rich, partially observable FPS settings. As a result, A2C completes a balanced trio of agents that span the classical value-based family, a modern exploration-enhanced extension, and a policy-gradient baseline—making it an ideal candidate for comparative analysis.

VI. EXPERIMENTAL SETUP

All experiments were conducted in the ViZDoom Defend-the-Line scenario, a fast-paced first-person environment in which the agent must eliminate approaching enemies while managing aiming efficiency and reaction timing. The environment presents partial observability and visually dense combat scenes, making it an effective benchmark for evaluating pixel-based control policies. Each agent was trained for 10 epochs, with 3,000 learning steps per epoch, yielding a total training budget of 30,000 environment steps. During interaction, actions were executed with a frame-repeat of 4, requiring the agent to issue temporally extended commands—a common technique for stabilizing behavior and reducing action-selection overhead.

All models operated on identical visual inputs, downsampled to 30×45 grayscale resolution, and used a discounted return factor of 0.99. Value-based agents (DQN and NoisyNet-DQN) were trained with a batch size of 64, a replay memory capacity of 100,000 transitions, and an optimization rate of 2.5×10^{-4} . Gradients were clipped to a maximum norm of 5.0, and target-network synchronization occurred every 1,000 training steps. The NoisyNet variant introduced trainable parameter noise within its fully connected layers, while the baseline model relied on ϵ -greedy exploration.

A2C agent was trained using rollouts of length 10, generalized advantage estimation (GAE) with a λ of 0.95, and coefficient weights of 0.5 and 0.01 for value and entropy regularization, respectively. To encourage precision in combat, all agents incorporated a lightweight shaping mechanism: an accuracy bonus that scaled with the ratio of kills to shots (0.2 per accuracy unit), and a shot penalty of 0.01 per firing event. Multi-step returns were computed using a

default n-step horizon of 3, enabling richer temporal credit assignment while maintaining training stability. Together, these design choices ensured that all agents operated under identical environmental and computational constraints, facilitating fair empirical comparison across architectural families.

VII. RESULTS

TABLE 2. Model Comparison

Agent	Avg R	Avg Elim	Avg Shots	Avg Acc	Best R	Best Acc
Baseline	11.09	12.08	68.64	0.18	27	0.3
Noisy	10.96	11.91	75.01	0.16	26	0.26
A2C	10.26	11.21	37.89	0.32	27	0.78

The comparative evaluation reveals several notable distinctions in how the three agents—Baseline DQN, Noisy DQN, and A2C—learn and express tactical behavior in the Defend-the-Line scenario. Across roughly equal episode counts, the Baseline DQN achieved the highest average reward (11.09) and the highest average kill count (12.08), indicating that value-based TD learning with replay memory is effective at acquiring a steady, reward-maximizing policy. The Noisy DQN performed similarly in terms of reward (10.96) and kills (11.91), suggesting that parameter-space exploration did not substantially alter final task competence, though its slightly higher average shot count indicates less conservative firing behavior. Both DQN variants achieved nearly identical best-episode reward values (27 for Baseline, 26 for Noisy), showing comparable peak performance ceilings.

In contrast, the A2C agent demonstrated a qualitatively different behavioral profile. Although its average reward (10.26) and average kills (11.21) were marginally lower than the DQN agents, A2C exhibited a dramatically lower average shot count (37.89 versus 68.64–75.00) and consequently the highest average accuracy (0.324), more than double that of the Noisy DQN. Its best-episode accuracy (0.778) further indicates that, when successful, A2C adopts a far more deliberate and resource-efficient combat style. This aligns with expectations for actor-critic methods, which typically produce smoother, more calibrated action probabilities and can internalize firing discipline more effectively than ϵ -greedy or noisy value-based methods.

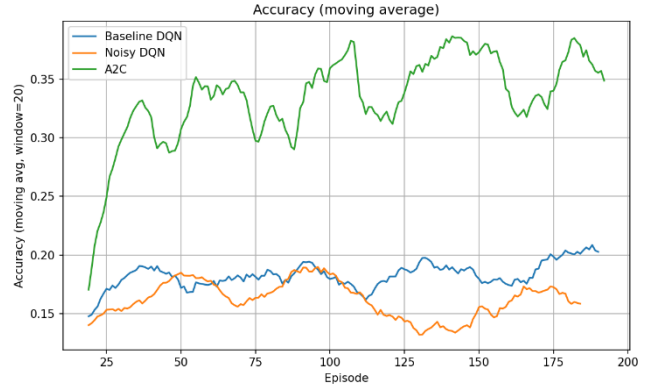


Fig 1. Aiming accuracy comparison across models

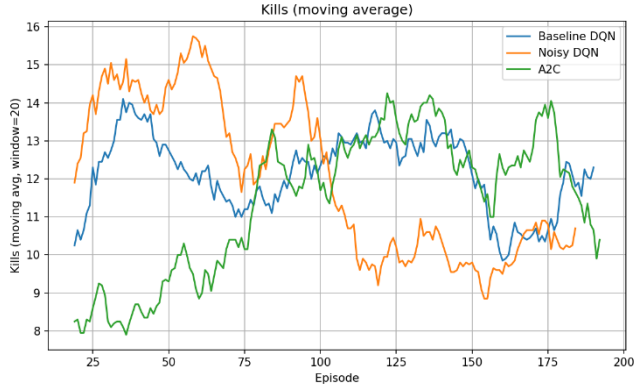


Fig 2 Elimination comparison across models

A deeper examination of the training logs and episode recordings indicates that the value-based methods—both Baseline DQN and Noisy DQN—tend to prioritize reward maximization through high-volume attack strategies, often at the expense of precision. These agents frequently exhibit overfitting to the training instance of Defend-the-Line, learning policies that rely on continuous or nearly continuous firing. In many recordings, the value-based models execute shots immediately after making small angular adjustments, even in the absence of visible enemies. This behavior is consistent with the reward structure: because kills provide substantial positive reinforcement while shots incur only modest penalties, the Q-learning agents implicitly favor aggressive firing policies that maximize the probability of incidental enemy hits.

In contrast, A2C demonstrates significantly more deliberate and context-aware firing behavior. Rather than saturating the action stream with attacks, A2C selectively targets enemies—often prioritizing nearer opponents to increase hit likelihood and reduce incoming damage. This pattern emerges naturally from the actor-critic formulation, in which the policy directly optimizes action probabilities rather than value estimates alone. As a result, A2C internalizes a firing discipline that reflects a more balanced trade-off between offensive and defensive considerations, leading to substantially higher accuracy and more human-like tactical decisions.

A. Ablation Study

To evaluate the impact of reward shaping on agent performance, an ablation study was conducted in which the accuracy bonus and shot-penalty terms were selectively disabled while keeping all other training conditions fixed. This manipulation isolates whether shaped rewards meaningfully influence learning dynamics beyond the raw elimination-based reward signal.

TABLE 3. Model Comparison Without Shaping

Agent	Avg R	Avg Elim	Avg Shots	Avg Acc	Best R	Best Acc
Baseline	16.48	17.45	101.35	0.17	35	0.29
Noisy	17.15	18.12	115.46	0.16	33	0.26
A2C	15.20	16.17	150.85	0.11	31	0.17

The ablation results reveal a clear shift in agent behavior when accuracy-based reward shaping is removed, with all models exhibiting more aggressive but less efficient firing

policies. Both value-based agents—Baseline DQN and NoisyNet-DQN—show increased average rewards (16.48 and 17.15, respectively) relative to the shaped-reward condition, driven largely by higher kill counts (17.45 and 18.12). However, this gain is accompanied by a substantial increase in average shot volume (101.35 and 115.46), resulting in reduced firing accuracy (0.17 and 0.16). This pattern is consistent with the intrinsic bias of Q-learning: in the absence of shaping, the agents optimize exclusively for kill-based rewards and thus adopt high-frequency firing strategies that maximize contact probability rather than precision. The NoisyNet variant exhibits slightly higher rewards and kills, suggesting that parameter-space exploration amplifies the drive toward aggressive firing when no mechanism penalizes inefficiency.

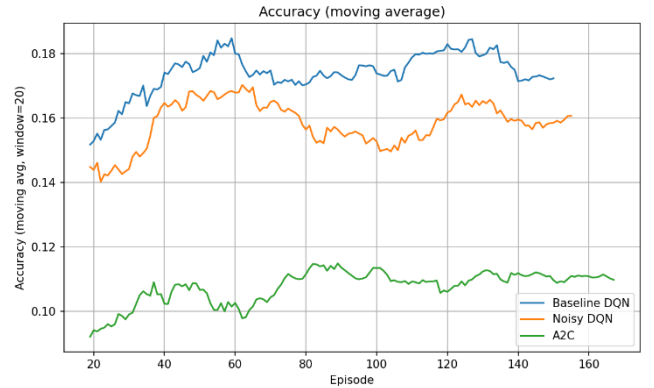


Fig 3 Aiming accuracy comparison across models without reward shaping

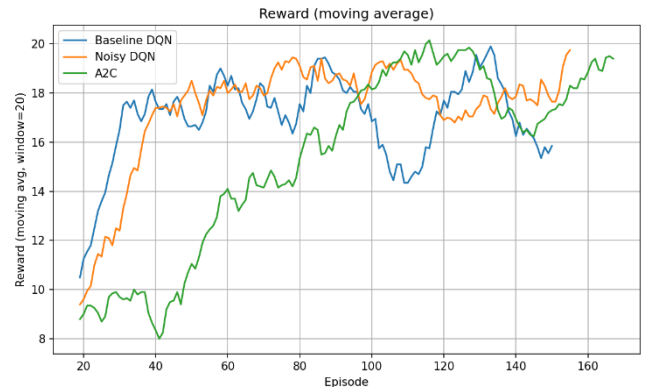


Fig 4 Reward comparison across models

The A2C model demonstrates a different but equally telling response to the removal of accuracy shaping. While its average reward (15.20) and kill count (16.17) remain competitive, A2C fires dramatically more shots on average (150.85), leading to the lowest overall accuracy (0.11). This indicates that without structured feedback encouraging disciplined shot selection, the policy-gradient method collapses toward a broad, exploratory firing strategy—prioritizing coverage over selectivity. A2C’s best-episode performance (31 reward, 0.167 accuracy) further reinforces the observation that, in unshaped settings, the actor-critic approach struggles to internalize firing discipline and instead leans toward policies that distribute probability mass across high-frequency attack actions.

Overall, these findings confirm that reward shaping plays a crucial role in regulating tactical behavior. Without accuracy incentives or penalties for excessive firing, all agents—especially A2C—overproduce shots and rely on brute-force

firing strategies to accumulate kills. Although this behavior yields higher raw reward in certain cases, it reflects a degradation in tactical quality and reveals a strong dependence on shaping signals to achieve balanced, human-like policies. This contrasts sharply with the shaped-reward condition, where A2C exhibited the highest firing discipline and the DQN variants displayed more stable, accurate policies. These ablation results therefore highlight the importance of shaping terms in guiding agents away from degenerate, high-volume attack patterns and toward behavior that aligns with the intended task dynamics.

VIII. DISCUSSION

The comparative evaluation reveals a clear trade-off between reward maximization and firing efficiency across agents. When reward shaping was removed, all models achieved higher average reward, but this gain was accompanied by a substantial increase in shot volume and a corresponding decrease in accuracy, indicating that the reward improvements were driven by aggressive firing rather than improved tactical performance. The value-based methods—Baseline DQN and NoisyNet-DQN—benefited most from the unshaped reward signal, achieving the highest raw rewards, but their policies became less stable and more prone to overfitting, frequently defaulting to high-frequency shooting with limited situational awareness.

Notably, although the A2C agent displayed the most coherent qualitative behavior under shaped conditions, it exhibited the lowest accuracy of all agents when reward shaping was removed, firing substantially more shots and failing to regulate its attack frequency. This decline reflects the sensitivity of policy-gradient methods to the reward landscape: without structured feedback, A2C’s stochastic policy collapses toward broad, undisciplined firing patterns. However, qualitative review of gameplay footage shows that under shaped rewards, A2C is capable of prioritizing nearer enemies, adjusting aim more deliberately, and exhibiting more human-like tactical judgment than its value-based counterparts. Together, these findings illustrate that while value-based agents maximize unshaped reward effectively, A2C offers more principled control and tactical nuance when guided by shaping signals.

IX. CONCLUSION

This study conducted a comparison of three reinforcement learning agents—Baseline DQN, NoisyNet-DQN, and A2C—within the ViZDoom Defend-the-Line scenario, focusing on how architectural choices, exploration strategies, and reward shaping mechanisms influence tactical performance in a visually complex, partially observable first-person environment. The results demonstrate that value-based methods are highly effective at maximizing cumulative reward and achieving strong kill counts, but tend to do so through high-volume firing strategies that sacrifice precision. Both the Baseline and NoisyNet-DQN agents frequently converged to behaviors characterized by rapid, repetitive shooting, often independent of enemy proximity. This reflects the inherent bias of Q-learning toward reward maximization via aggressive action repetition, especially when shaping signals are removed or weakened.

In contrast, the A2C agent exhibited a markedly different behavioral profile. Under shaped reward conditions, A2C demonstrated superior firing discipline and substantially higher accuracy, selectively engaging enemies and

prioritizing shots that had a higher probability of success. Although A2C underperformed in raw reward compared to DQN variants in the unshaped condition, its sensitivity to shaping and its more coherent tactical behavior underscore the strengths of actor–critic optimization in managing ambiguous or delayed visual feedback. The ablation experiments further revealed that A2C’s policy structure is particularly well suited for incorporating auxiliary signals, such as accuracy bonuses or penalties, which enable the agent to internalize higher-level tactical objectives beyond simple kill accumulation.

Taken together, these findings indicate that the choice of reinforcement learning architecture has a profound impact on emergent behavior in FPS environments. While value-based agents excel at aggressive, reward-maximizing strategies, actor–critic agents like A2C show promise for developing more precise, resource-efficient combat policies that better align with intended task objectives. A2C’s flexibility, responsiveness to reward shaping, and potential for integration with advanced exploration strategies or perceptual modules position it as a strong candidate for future experimentation. Continued work may explore extensions such as PPO, IMPALA, hierarchical actor–critics, or transformer-based perception layers to further enhance situational awareness, generalization, and robustness in complex combat scenarios.

X. FUTURE WORKS

While the present study focuses on marksmanship and tactical decision-making within the constrained Defend-the-Line scenario, achieving full autonomy in ViZDoom or similar FPS environments requires agents to master a far broader set of skills—including navigation, map exploration, resource management, and survival behaviors such as health collection and threat avoidance. These tasks vary substantially in structure and difficulty, suggesting that no single reinforcement learning architecture is likely to excel uniformly across them. The results of this study highlight that value-based methods are well suited for high-frequency, reward-driven combat engagement, whereas actor–critic methods demonstrate strengths in precision, selective action, and sensitivity to shaping signals. Leveraging these complementary capabilities motivates a more holistic approach in which multiple specialized models operate collaboratively, each addressing a specific aspect of the overall gameplay loop.

Such a modular framework could, for example, employ a navigation-focused agent—potentially using recurrent or transformer-based architectures—to map out environments and plan movement trajectories, while a combat module powered by A2C or an advanced actor–critic variant manages target engagement and firing discipline. Additional sub-agents could handle tasks such as resource gathering, evasive maneuvers, or long-term strategy, coordinating through hierarchical or multi-agent reinforcement learning mechanisms. This integrated approach mirrors how complex human or robotic systems decompose behavior into coordinated subtasks and would enable RL agents to scale from isolated tactical proficiency toward full-spectrum gameplay competence. As research progresses, combining specialized architectures, curriculum learning, and shared representation learning promises a viable path toward robust, general-purpose agents capable of operating effectively across the diverse challenges posed by modern FPS environments.

XI. REFERENCES

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